

Agentic RAG

End-to-end walkthrough of the **canonical production pipeline**: clients
→ FastAPI → runner → canonical graph v1 → verification → response.

Written from live code paths in `src/`.

What this document is for. Keep it beside the repo. Every major step names the function, file, and decision. If code and this manual diverge, trust the code — then update this file. Companion index: `docs/DOCUMENTATION_MAP.md`.

Edition: September 2026 (canonical v1)

Pipeline version: `canonical-v1`

Default mode: `canonical`

Canonical entry: `run_agent()` → `ask_canonical()`

Stack: LangGraph · LangChain · FastAPI · Chroma · OpenAI · Redis · React

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1. How to read this document

Start with §2–§4 for the whole picture. §6 is the canonical graph (replaces the retired eight-mode multi-graph runtime). §8–§9 cover retrieval and verification. §10–§11 for ops.

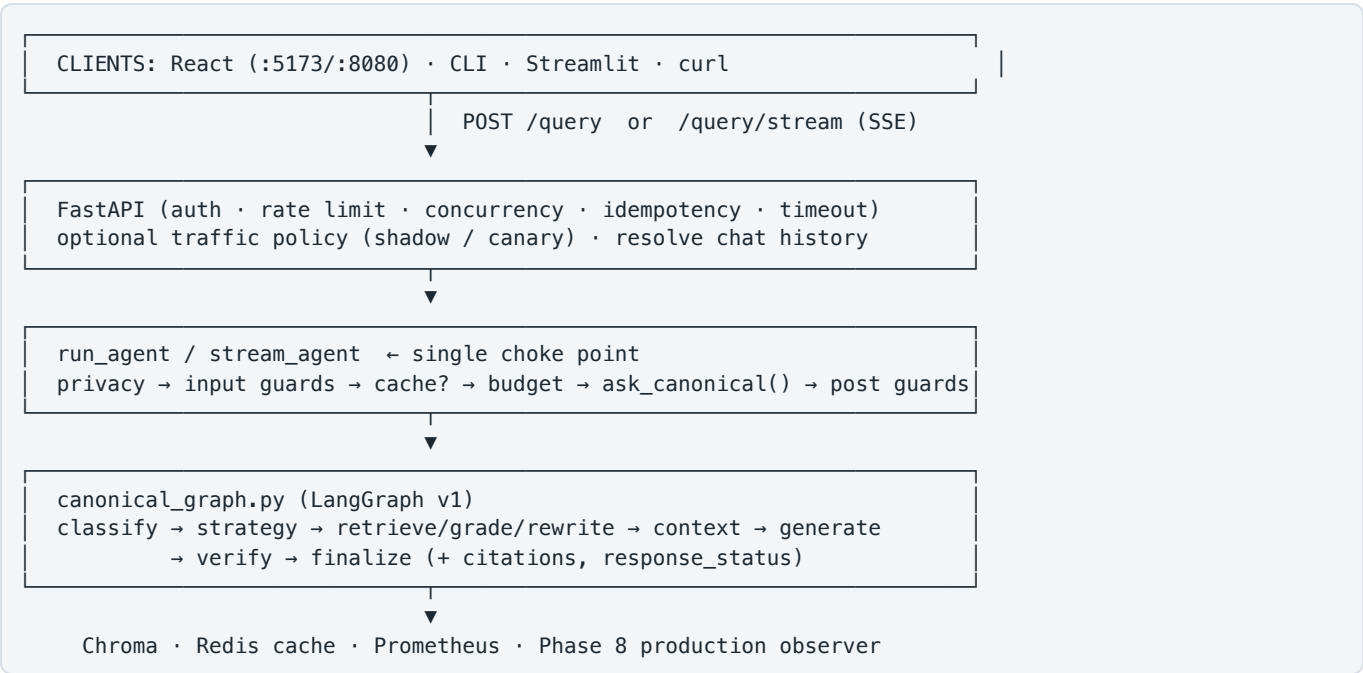
If you need...	Go to
One diagram of the whole product	§2 System at a glance
Exact steps on every API call	§4 Request lifecycle
Graph nodes, edges, state	§6 Canonical graph
What is an agent vs a function	§7 Agent boundaries
PDF → Chroma → hybrid retrieve	§8 Ingestion & retrieval
Verification & abstention	§9 Verification
Shadow/canary/continuous eval	§10 Phase 8

HISTORICAL NOTE

Phases 1–7 originally built eight separate mode graphs (`router_graph` , `crag_graph` , ...). Those modules were **removed** in Phase 7. Deprecated mode strings still accepted by the API map to canonical strategies internally. See `docs/ROADMAP.md` and `docs/LEGACY_RETIREMENT_INVENTORY.md` .

2. System at a glance

Agentic RAG is a **single canonical LangGraph workflow** with strategy selection, evidence grading, verification, and production controls (auth, cache, guardrails, continuous evaluation). One question flows through `classify` → `retrieve` (or `direct/web`) → `generate` → `verify` → `finalize`.



Mode key	Status	Behavior
canonical	Production	Full canonical graph; strategy selected inside graph.
baseline, router, crag, ...	Deprecated	Accepted for compatibility; mapped to force_strategy; deprecation metrics emitted.

3. Layered architecture

Layer	Responsibility	Primary modules
1. Interface	HTTP, CLI, UI, SSE	src/api/server.py, documents.py, frontend/
2. Control plane	Privacy, guards, cache, memory, budgets	runner.py, guardrails.py, cache/
3. Canonical graph	LangGraph workflow + verification	graph/canonical_graph.py, contracts/
4. Knowledge plane	Ingest, hybrid retrieve, rerank, federation	ingestion/, retrieval/, sources/
5. Model & tools	LLM, chains, safe web fetch	llm.py, chains/, tools/safe_web.py
6. Evaluation	Shadow, canary, continuous eval, gates	evaluation/, /ops/*

HARD RULE

Never call ask_canonical() or graph nodes from a new interface directly. Always go through run_agent() or stream_agent().

4. End-to-end request lifecycle

4.1 Master flow

```
Client
→ middleware (RequestId, CORS, TrustedHost?)
→ verify_api_key + enforce_client_rate_limit
→ (sync) Idempotency-Key lookup
→ _resolve_chat_history
→ concurrency semaphore
→ run_agent / stream_agent
  A. PrivacyGuard.apply_input
  B. InputGuardrails.validate (+ injection scan)
  C. should_use_cache?
  D. augment_question_with_history
  E. cache hit? → post-guards → return
  F. CostGuardrails + record_query
  G. _dispatch → ask_canonical()
  H. OutputGuardrails + privacy output scan
  I. optional QualityGuardrails
  J. generate_follow_ups
  K. set_cached_response (pipeline_version keyed)
  L. production_observer hook (Phase 8)
→ _persist_exchange (Supabase if configured)
→ QueryResponse / SSE
```

4.2 Inside _dispatch

File: `src/runner.py`. All modes route to the canonical graph:

```
normalized = normalize_mode(mode)
strategy = resolve_canonical_strategy(normalized) # deprecated aliases → strategy
canonical = ask_canonical(question, force_strategy=strategy)
return canonical_to_api_response(canonical, display_mode="canonical")
```

On canonical failure with `legacy_runtime_enabled=false` (default): safe abstention response — no legacy graph replay.

4.3 Streaming — POST /query/stream

SSE <code>event</code>	Meaning
<code>step</code>	Graph progress (classify, retrieve, grade, ...)
<code>pipeline</code>	Structured debug payload for frontend pipeline panel
<code>token</code>	Answer token stream
<code>answer</code>	Full answer after generation
<code>sources</code>	Source labels + citations array
<code>follow_ups</code>	Grounded follow-up questions
<code>done</code>	Terminal: latency, session, verification fields, pipeline_version
<code>error</code>	Guardrail or runtime failure

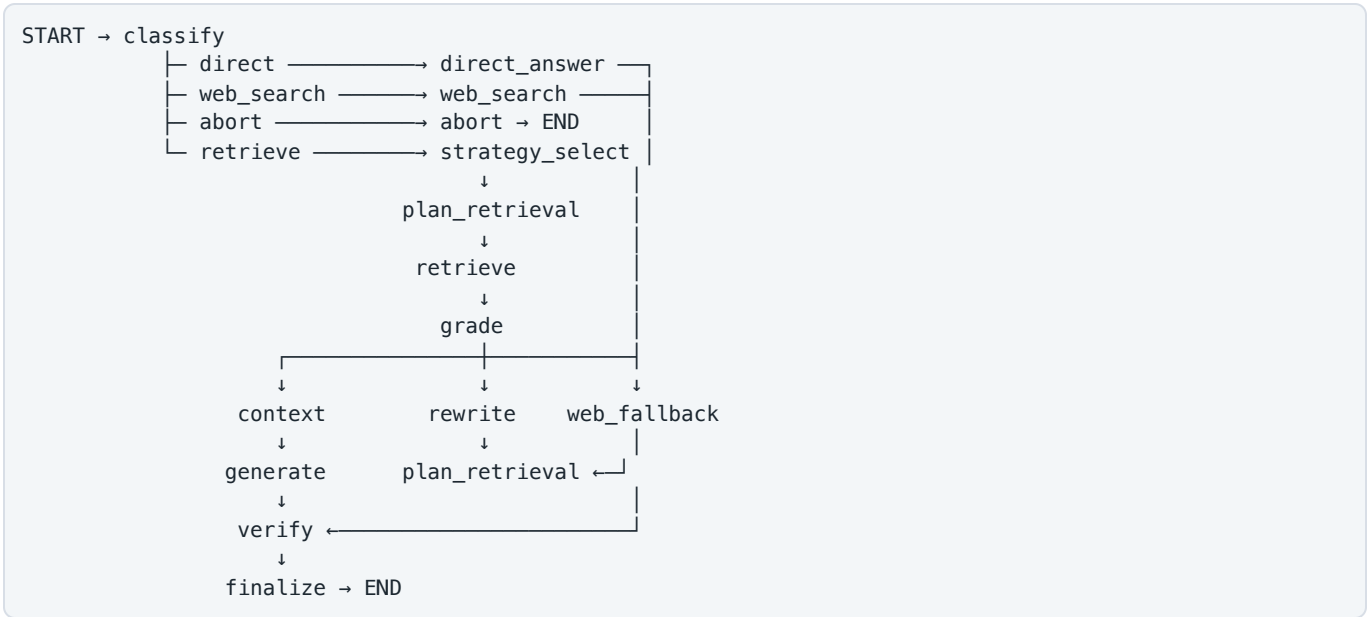
5. Conversation memory

Memory packing in `src/memory/chat_memory.py` ; optional Supabase persistence in `supabase_store.py` .

1. Recent N exchanges: full Q+A (truncated).
2. Older exchanges: questions only (topic trail).
3. Wrapped into `augment_question_with_history()` before `ask_canonical` .
4. Cache skipped when memory + non-empty history (answers must not leak across sessions).

6. Canonical LangGraph (production)

File: `src/graph/canonical_graph.py` · **State:** `CanonicalAgentState (src/contracts/state.py)` ·
Builder: `build_canonical_graph()`



Node	Kind	Module
<code>classify</code>	LLM router	<code>shared_nodes.invoke_router</code>
<code>strategy_select</code>	Heuristic + LLM	<code>strategy_heuristics.select_strategy</code>
<code>plan_retrieval / retrieve</code>	Deterministic	<code>retrieval/canonical_adapter.py</code>
<code>grade</code>	LLM grader	<code>evidence_management.py</code>
<code>rewrite</code>	LLM rewriter	<code>agents/query_rewriter.py</code>
<code>context</code>	Deterministic	<code>context_builder.py</code>
<code>generate</code>	LLM	<code>chains/generation.rag_chain</code>
<code>verify</code>	Rules + optional LLM	<code>verification_engine.py</code>
<code>finalize</code>	Deterministic	<code>citation_mapper.py</code> , API mapping

Strategies (internal, not separate graphs): `simple` , `decompose` , `multi_hop` . Deprecated API mode strings can force a strategy via `runner_modes.resolve_canonical_strategy` .

7. LLM agents vs deterministic functions

LLM decision components	Deterministic pipeline
Router (<code>classify_query</code>)	Hybrid dense + BM25 + RRF fusion
Strategy selector	Retrieval plan execution
Document grader	Cross-encoder rerank (NVIDIA / FlashRank)
Query rewriter	Parent-child expansion, deduplication
Generator	Token budgeting, citation index mapping
Verifier (optional LLM judge)	SSRF-safe web fetch, RBAC filters

NOT IN PRODUCTION GRAPH

In-graph ReAct tool loop and seven separate mode graphs were removed in Phase 7. `@tool` wrappers remain for federation tests and deterministic helpers — not LLM-selectable in the canonical graph.

8. Ingestion & retrieval

8.1 Ingestion

```
python -m src.ingestion.ingest --source data/sample_docs
→ cleanse → section_parent_child chunking → embed → Chroma upsert
→ parent_store.json → invalidate BM25 + flush cache
```

Async jobs: `POST /ingest/jobs`, upload: `POST /ingest/upload`. Registry: `document_registry.py`.

8.2 Retrieval (deterministic)

```
query
→ over-fetch candidate_k (hybrid: dense + BM25 → RRF)
→ optional rerank (nvidia API or flashrank local)
→ RBAC tenant/role filters
→ optional federation (SQLite / ops API / lab MCP)
→ parent expansion → evidence candidates for grading
```

Entry: `execute_retrieval_plan()` in `src/retrieval/canonical_adapter.py`.

9. Generation, verification, citations, streaming

- **Generation:** LCEL chains in `src/chains/generation.py`; streaming via `stream_text`.
- **Verification:** Always runs before finalize. Outcomes: `PASS`, `WARN`, `FAIL`. May abstain or append caveat.
- **Citations:** `citation_mapper.map_citations` links evidence IDs to user-facing `Citation` objects.
- **Response status:** `answered` | `answered_with_warning` | `abstained`.
- **API bridge:** `canonical_to_api_response()` in `src/contracts/api_response.py`.
- **Follow-ups:** `generate_follow_ups` attached by runner after successful non-abort runs.

10. Production envelope & Phase 8 evaluation

Concern	Where
Auth + rate limits	<code>api/security.py</code> , <code>api/rate_limit.py</code>
Injection / PII	<code>security/injection.py</code> , <code>privacy.py</code>
Exact + semantic cache	<code>cache/redis_cache.py</code> , <code>semantic_cache_redis.py</code> (pipeline-version keyed)
Shadow / canary	<code>evaluation/shadow_*.py</code> , <code>canary_rollout.py</code>
Continuous eval	<code>evaluation/continuous_eval.py</code> , <code>eval_gates.py</code>
Quality dashboard	<code>GET /ops/quality/dashboard</code>
Legacy retirement status	<code>GET /ops/legacy/retirement</code>
Feedback loop	<code>POST /feedback</code> , <code>feedback_classifier.py</code>
Metrics	<code>GET /metrics</code> — includes canonical, canary, deprecated-mode counters

Defaults: `CANONICAL_PRIMARY=true`, `LEGACY_RUNTIME_ENABLED=false`, `CONTINUOUS_EVAL_ENABLED=true`.

11. API surface & deployment

Method	Path	Purpose
GET	/health , /health/ready	Liveness / readiness
GET	/metrics	Prometheus
POST	/query , /query/stream	Sync / SSE query
GET	/modes , /config	Mode list + server config
GET/DELETE	/documents	Index management
POST	/ingest/upload , /ingest/jobs	Ingestion
POST	/feedback	User feedback
GET	/ops/preflight , /ops/canary/* , /ops/eval/*	Operator tools

```
docker compose up -d          # API :8000, frontend :8080, Redis, Chroma
uvicorn src.api.server:app --reload --port 8000  # local dev
```

12. Configuration map

All settings: `src/config.py` via pydantic-settings. Key groups:

Group	Examples
Canonical	CANONICAL_PRIMARY , LEGACY_RUNTIME_ENABLED , canonical_verification_llm_enabled
Retrieval	RETRIEVAL_SEARCH_TYPE , RETRIEVAL_TOP_K , RERANK_PROVIDER
Security	API_KEY , REQUIRE_API_KEY , TRUST_CLIENT_RBAC
Cache	CACHE_ENABLED , REDIS_URL , semantic cache thresholds
Eval / canary	SHADOW_ENABLED , CANARY_ENABLED , CONTINUOUS_EVAL_ENABLED
Memory	MEMORY_* , SUPABASE_*

13. File ownership quick reference

Path	Owns
src/runner.py	Choke point: guards, cache, dispatch to canonical
src/graph/canonical_graph.py	Production LangGraph
src/contracts/	State, evidence, verification, API response mapping
src/retrieval/canonical_adapter.py	Deterministic retrieval boundary
src/evaluation/	Shadow, canary, continuous eval, regression gates
src/api/server.py	HTTP routes, SSE, ops endpoints
frontend/	Chat UI, pipeline debug panel, documents, feedback

14. Known gaps

Item	Reality
Idempotency	Sync /query only, not stream
Quality guardrails	Off by default (extra LLM cost)
CLI / Streamlit	Bypass HTTP auth, rate limits, concurrency
BM25 index	Per-process; not shared across API workers
Benchmark numbers	Not published in docs — run evaluators locally

15. Tests that document behavior

File	Documents
tests/test_canonical_graph.py	Graph topology, nodes, verification path
tests/test_canonical_production.py	Runner dispatch, API response fields
tests/test_phase8_continuous_eval.py	Continuous eval loop, dashboard
tests/test_shadow_evaluation.py	Shadow paired comparison
tests/test_documentation_consistency.py	Doc ↔ code alignment
tests/test_production_safety.py	Auth, capacity, rate limits

The repository maintains an extensive automated test suite — do not rely on hardcoded counts in documentation.

16. Worked traces

```
# Canonical (preferred)
python -m src.cli ask "What is Self-RAG?" --mode canonical -v

# Comparison / strategy selection visible in steps
python -m src.cli ask "Compare naive RAG and advanced RAG" --mode canonical -v

# HTTP
curl -s -X POST http://localhost:8000/query \
  -H "Content-Type: application/json" \
  -d '{"question":"What is Self-RAG?","mode":"canonical"}' | jq .

# Streaming + pipeline debug events
curl -N http://localhost:8000/query/stream \
  -H "Content-Type: application/json" \
  -d '{"question":"What is Self-RAG?","mode":"canonical"}'

# Phase 8 ops (requires API key when auth enabled)
curl -H "X-API-Key: $API_KEY" http://localhost:8000/ops/quality/dashboard
```

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Authoritative docs: [docs/DOCUMENTATION_MAP.md](#), [docs/ARCHITECTURE.md](#), [docs/LANGGRAPH.md](#), [docs/API.md](#), [docs/EVALUATION.md](#).

If code and this manual diverge, trust the code — then update this file.